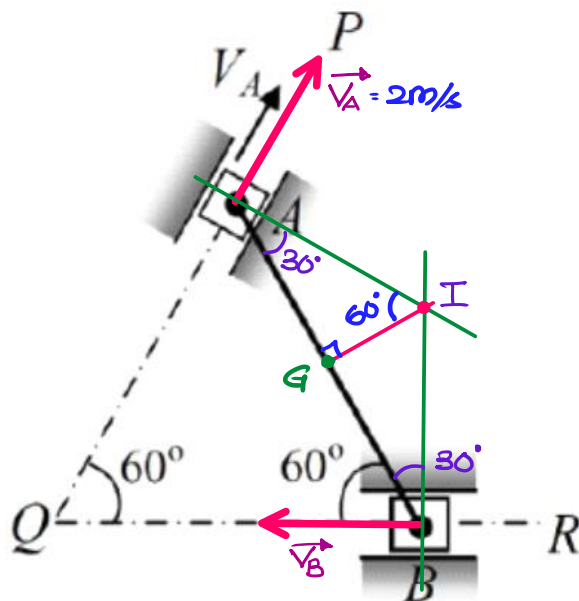
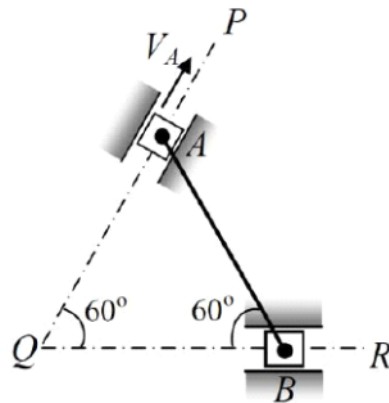


2M.

The rod AB , of length 1 m, shown in the figure is connected to two sliders at each end through pins. The sliders can slide along QP and QR . If the velocity V_A of the slider at A is 2 m/s, the velocity of the midpoint of the rod at this instant is _____ m/s.



$$V_G = \{IG\} \omega ; V_A = \{IA\} \omega$$

$$\Rightarrow \frac{V_G}{IG} = \frac{V_A}{IA}$$

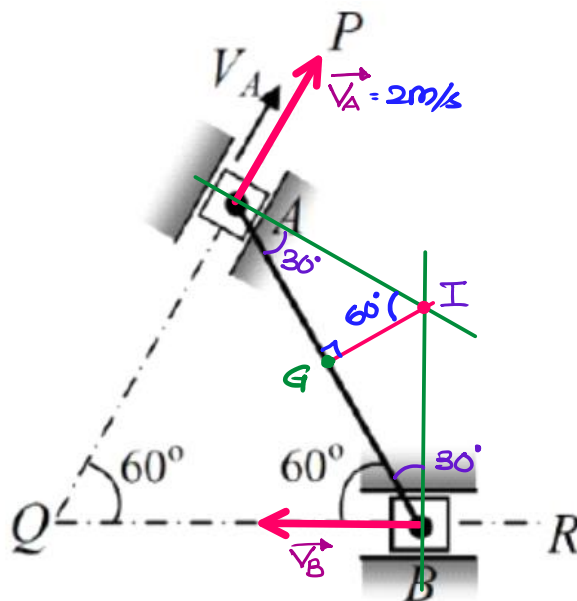
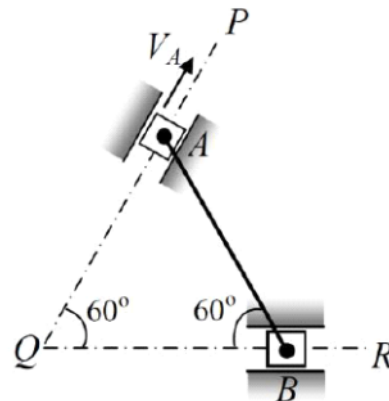
$$\Rightarrow V_G = V_A \left\{ \frac{IG}{IA} \right\}$$

$$= V_A \left(\frac{IA \cos 60^\circ}{IA} \right)$$

$$\therefore V_G = 1 \text{ m/s}$$

2M.

The rod AB , of length 1 m, shown in the figure is connected to two sliders at each end through pins. The sliders can slide along QP and QR . If the velocity V_A of the slider at A is 2 m/s, the velocity of the midpoint of the rod at this instant is _____ m/s.



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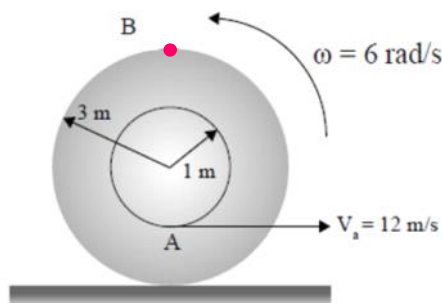
$$\Rightarrow \frac{V_G}{IG} = \frac{V_A}{IA}$$

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$$\therefore V_G = 1 \text{ m/s}$$

The compound disk in the figure rolls on the horizontal floor. The linear velocity of the cord leaving A is 12 m/s rightward and the disk rotates counter-clockwise at 6 rad/s. The total velocity of the point B is



- (a) 4 m/s
(c) 16 m/s

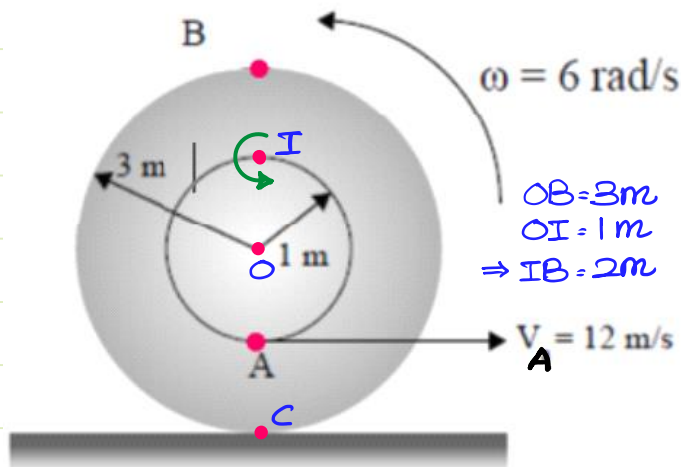
- (b) 12 m/s
(d) 20 m/s

Questions :

1. $\vec{V}_B = ?$

2. Identify if the disc is rolling with/ without slip.

3. If it is rolling with slip find V_{slip} .



$$V_A = R_{A/I} \cdot \omega$$

$$12 = R_{A/I} \{6\} \Rightarrow R_{A/I} = 2m$$

There are 2 possibilities :

1. I Centre lies 2m above A.
2. I Centre lies 2m below A.

If I Centre lies 2m below A: $\Rightarrow V_A$ is in -ve x axis

But V_A is given in +ve x axis $\therefore$ I Centre lies 2m above A.
 $\therefore$ I Centre is not at pt. of contact $\therefore$ Rolling with slip.

$$V_B = R_{B/I} \cdot \omega$$

$$= 2m \times 6 \text{ rad/s}$$

$$\therefore V_B = 12 \text{ m/s} \leftarrow$$

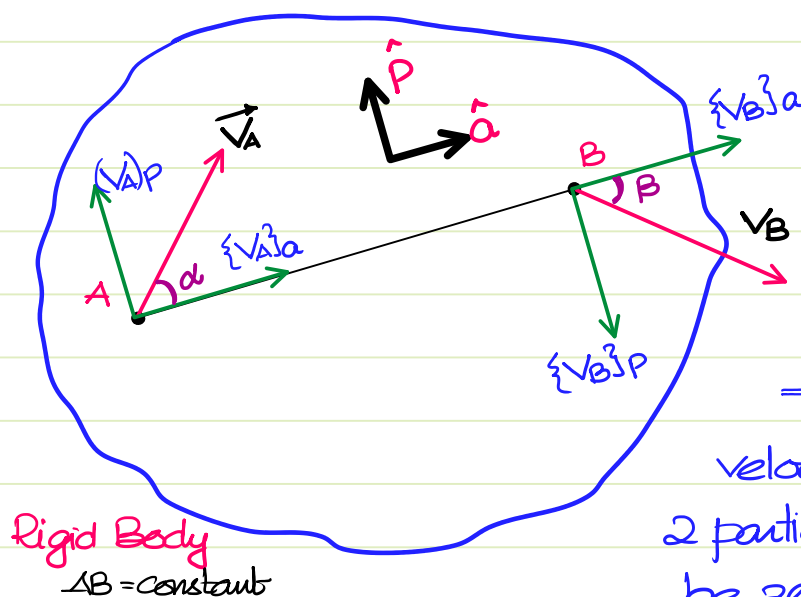
V_{slip} = Relative Velocity at pt. of Contact
 $\Rightarrow V_{\text{slip}} = V_{C/\text{surface}} = \vec{V}_C - \vec{V}_{\text{surface}}$

$$\therefore V_{\text{slip}} = V_C = R_{C/I} \cdot \omega$$

$$= 4 \times 6$$

$$\therefore V_{\text{slip}} = 24 \text{ m/s} \rightarrow$$

** Relative Velocity Method :



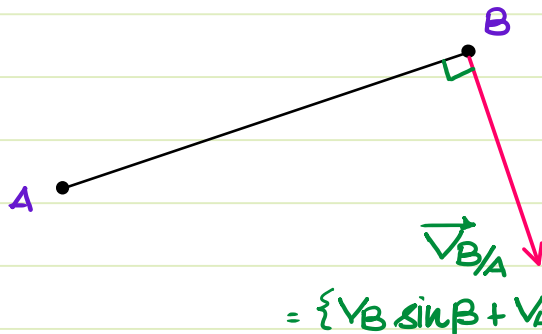
Rigid Body
 $AB = \text{constant}$

$\therefore$ the body is rigid,
 $\therefore |AB| = \text{constant}$
 $\Rightarrow (V_A)_{\text{along } AB} = (V_B)_{\text{along } AB}$

$$V_A \cos \alpha = V_B \cos \beta$$

$\Rightarrow$ The component of relative velocity along the line joining 2 particles in a rigid body must be zero.

⇒ The relative velocity must have only one component in the dirⁿ $\perp$ to the line joining 2 particles.



$$\vec{V}_B = (V_B \cos \beta) \hat{a} - (V_B \sin \beta) \hat{p}$$

$$\vec{V}_A = (V_A \cos \alpha) \hat{a} + (V_A \sin \alpha) \hat{p}$$

$$\vec{V}_{B/A} = \{V_B \sin \beta + V_A \sin \alpha\}$$

$$\vec{V}_{B/A} = \vec{V}_B - \vec{V}_A$$

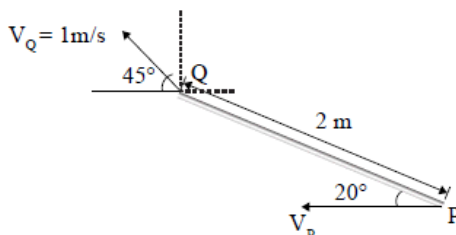
$$= \{V_B \cos \beta - V_A \sin \alpha\} \hat{a} + (-V_B \sin \beta - V_A \sin \alpha) \hat{p}$$

$$V_{B/A} = r_{B/A} \cdot \omega$$

Tips for solving problems:

1. Rel. Velocity along link = 0 $\Rightarrow V_B \cos \beta = V_A \cos \alpha$.
2. for angular velocity : $\omega = \frac{V_{B/A}}{r_{B/A}}$.

11. A rigid link PQ is 2 m long and oriented at 20° to the horizontal as shown in the figure below. The magnitude and direction of velocity V_Q and the direction velocity V_P are given. The magnitude of V_P (in m/s) at this instant is: (GATE-2014)



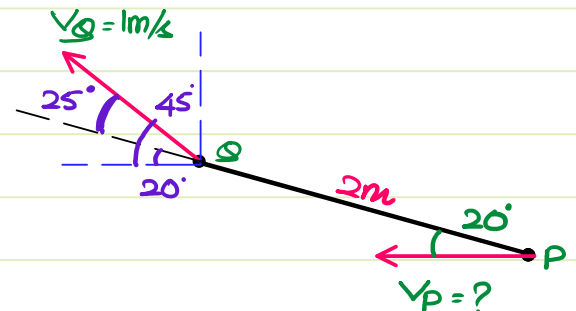
- (a) 2.14
(b) 1.89
(c) 1.21
(d) 0.96

Find ω by Rel. Vel. Method.

$V_{P/Q}$ is $\perp$ to PQ

$$V_{P/Q} = V_P \sin 20^\circ + V_Q \sin 25^\circ = r_{P/Q} \cdot \omega$$

$$= 0.7525 = 2 \cdot \omega \Rightarrow \omega = 0.376 \text{ rad/s}$$



$\therefore PQ$ is rigid

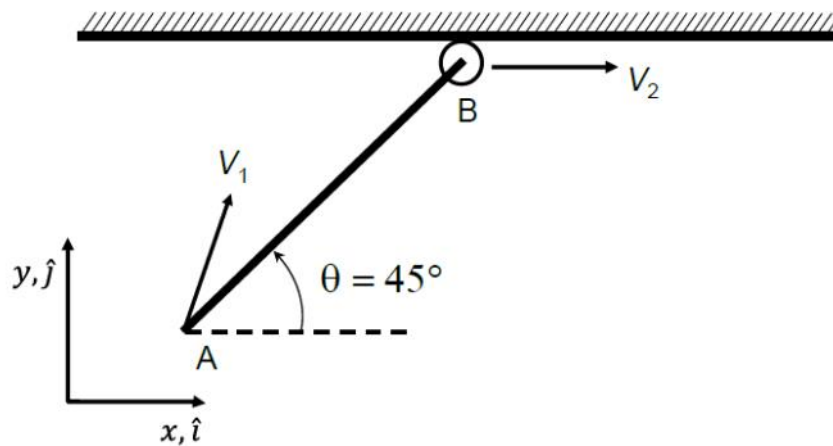
$\Rightarrow V_P$ along PQ = V_Q along PQ.

$$V_P \cos 20^\circ = V_Q \cos 25^\circ$$

$$\Rightarrow V_P = \frac{V_Q \cos 25^\circ}{\cos 20^\circ} = \frac{1 \times \cos 25^\circ}{\cos 20^\circ}$$

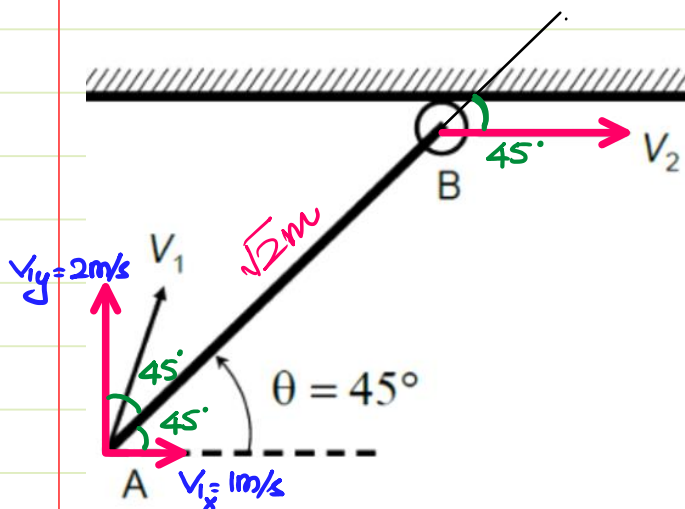
$$\therefore V_P = 0.9644 \text{ m/s}$$

A rigid rod (AB) of length $L = \sqrt{2}$ m is undergoing ^{GPM.} translational as well as rotational motion in the x-y plane (see the figure). The point A has the velocity $\underline{V}_1 = \hat{i} + 2\hat{j}$ m/s. The end B is constrained to move only along the x direction.



The magnitude of the velocity V_2 (in m/s) at the end B is _____

$\therefore AB$ is rigid, $\therefore V_A$ along $AB = V_B$ along AB .



$$V_{1x} \cos 45^\circ + V_{1y} \sin 45^\circ = V_2 \cos 45^\circ$$

$$\Rightarrow V_2 = V_{1x} + V_{1y}$$

$$= 1 + 2$$

$$\therefore \boxed{V_2 = 3 \text{ m/s}}$$

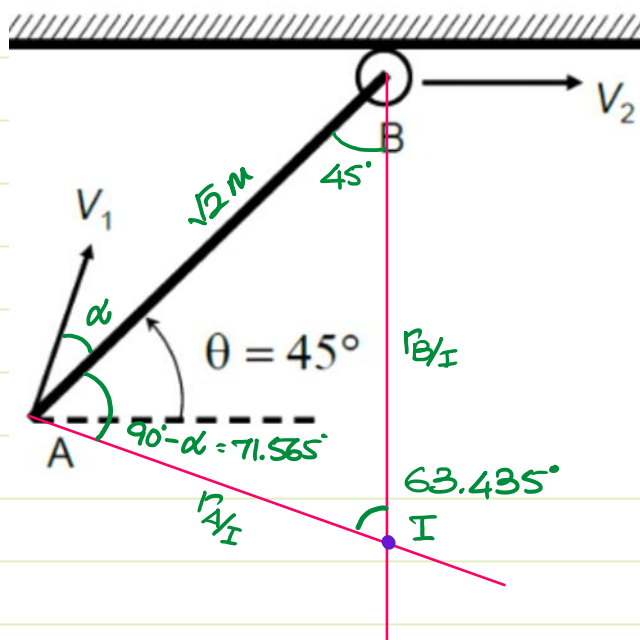
$$\underline{V}_B = \{3\hat{i} + 0\hat{j}\} \text{ m/s.}$$

$$\omega = ?$$

$$\omega = \frac{V_{AB}}{r_{AB}} = \frac{|\underline{V}_A - \underline{V}_B|}{\sqrt{2}} = \frac{|\hat{i} + 2\hat{j} - (3\hat{i} + 0\hat{j})|}{\sqrt{2}}$$

$$= \frac{|-2\hat{i} + 2\hat{j}|}{\sqrt{2}}$$

$$= \frac{2\sqrt{2}}{\sqrt{2}} \quad \therefore \boxed{\omega = 2 \text{ rad/s}}$$



$$\alpha = \tan^{-1} \left\{ \frac{V_{1y}}{V_{1x}} \right\} - 45^\circ$$

$$= \tan^{-1} 2 - 45^\circ$$

$$= 18.435^\circ$$

$$90^\circ - \alpha = 71.565^\circ$$

By sine rule

$$\frac{r_{A/I}}{\sin 45^\circ} = \frac{r_{B/I}}{\sin 71.565^\circ} = \frac{\sqrt{2}}{\sin 63.435^\circ}$$

$$r_{A/I} =$$

$$r_{B/I} =$$

$$\checkmark \checkmark$$

$$V_A = r_{A/I} \cdot \omega$$

$$\underline{V_B = r_{B/I} \cdot \omega}$$

4. Kinetics

4.1. Newtons' 2nd Law of Motion

4.1.1. Particle Study

* Newtons' 2nd Law of Motion

* D'Alemberts Principle.

* Centrifugal force

Newton's 2nd Law of Motion: When a non-zero unbalanced force acts on a body $\{\sum \vec{F} \neq 0\}$, then the body accelerates in the direction of unbalanced force with a Magnitude proportional to the force.

$$\sum \vec{F} \neq 0$$

$$\Rightarrow \sum \vec{F} \propto \vec{a}$$

$$\Rightarrow \boxed{\sum \vec{F} = m \cdot \vec{a}}$$

m : Mass of the body resists accⁿ, $\therefore$ it is the representative of Linear Inertia.

→ Newton's 2nd Law of Motion.

D'Alemberts' Principle: When a body experiences a non zero unbalanced force $\{\sum \vec{F} \neq 0\}$, then an imaginary force is considered in the dirⁿ opposite to that of acceleration in order to establish dynamic eqⁿ.

Note: This imaginary force is known as Inertia force & is always equal to $m \cdot a$.

By Newtons' 2nd Law: $\sum \vec{F} = m\vec{a}$

$$\Rightarrow \boxed{\sum \vec{F} + \{-m\vec{a}\} = 0} \quad \text{Dynamic Eqⁿ}$$

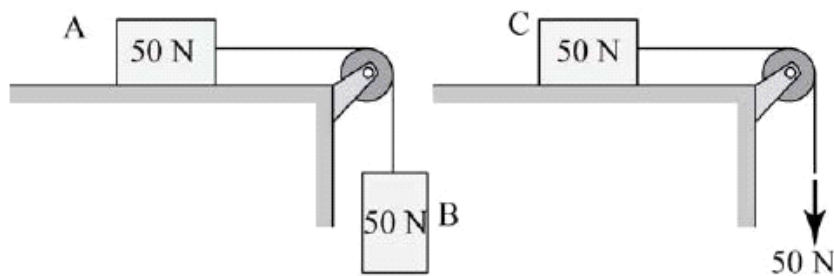
↓
Inertia force / Pseudoforce

If a body is undergoing a motion where the dirⁿ of $\vec{v}$ is changing then it must observe an accⁿ known as **Centripetal Accⁿ**.

The inertia force corresponding to Centripetal Accⁿ is known as **Centrifugal force** $\{ F_c = m r \omega^2 = m \frac{v^2}{r} \}$

F_c must be directed along the radius & away from the centre of curvature **{ Radially outward. }**

Two systems shown below start from rest. For the system shown on the left, two 50N blocks are connected by a cord. For the system shown on the right, the 50N block is pulled by a 50N downward force. Neglect friction. Which of the following is true?



- (A) Blocks A and C have the same acceleration.
~~(B)~~ Block C will have a larger acceleration than block A.
 (C) Block A will have a larger acceleration than block C.
 (D) Block A will not move.

Let a_1 be the accⁿ of block A & block B:

$$\sum F = m \cdot a_1$$

$$\Rightarrow 50 = \{ m_A + m_B \} a_1 \Rightarrow 50 = \left\{ \frac{50}{g} + \frac{50}{g} \right\} a_1$$

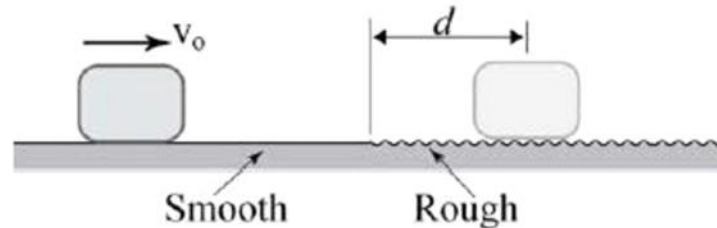
$$\Rightarrow a_1 = g/2$$

Let a_2 be the accⁿ of block C:

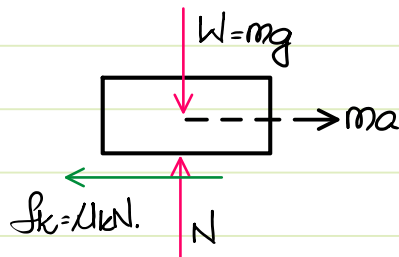
$$\sum F = m a_2$$

$$\Rightarrow 50 = m \cdot a_2 \Rightarrow 50 = \left\{ \frac{50}{g} \right\} a_2 \quad \therefore a_2 = g$$

- 2M. A block is travelling with a constant speed v_0 on a smooth surface when the surface suddenly becomes rough with a coefficient of friction μ , which causes the block to stop after a distance d . When the block travels twice as fast, i.e. at a speed $2v_0$ it travels a distance D on the rough surface before stopping. Find the ratio D/d .



Find the retardation on the rough surface:



$$\sum F_y = 0 : N = mg \Rightarrow f_k = \mu mg$$

$$\sum F_x = 0 : f_k = m \cdot a$$

$$\Rightarrow \mu mg = ma \quad \therefore \boxed{a = \mu g}$$

When a body retards on a rough horizontal surface, without any other horizontal force, then $a = \mu g$.

$$\therefore \text{Retardation is constant} \quad \therefore v^2 - u^2 = 2as$$

$$\Rightarrow s \propto \{v^2 - u^2\} \text{ if } a \text{ is same in both cases.}$$

$$\Rightarrow \frac{D}{d} = \frac{\{0^2 - (2v_0)^2\}}{\{0^2 - v_0^2\}} = \frac{-4v_0^2}{-v_0^2} \quad \therefore \boxed{\frac{D}{d} = 4}$$

- 1M. A child of mass **5 kg** is going round a merry-go-round that makes 1 rotation in **3.14s**. The radius of the merry-go-round is **2 m**. The centrifugal force on the child will be

- (a) 40 N (b) 50 N
(c) 80 N (d) 100 N

$$\omega = 2\pi/t = 2\pi/3.14$$

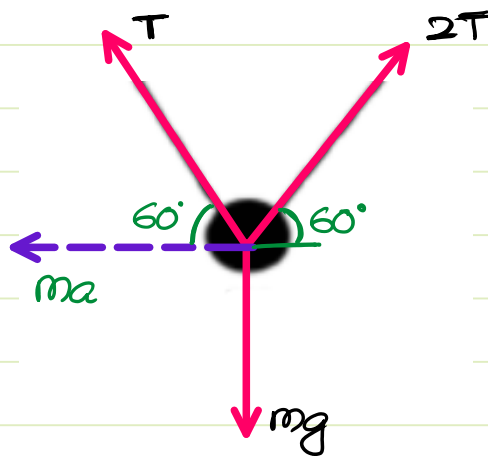
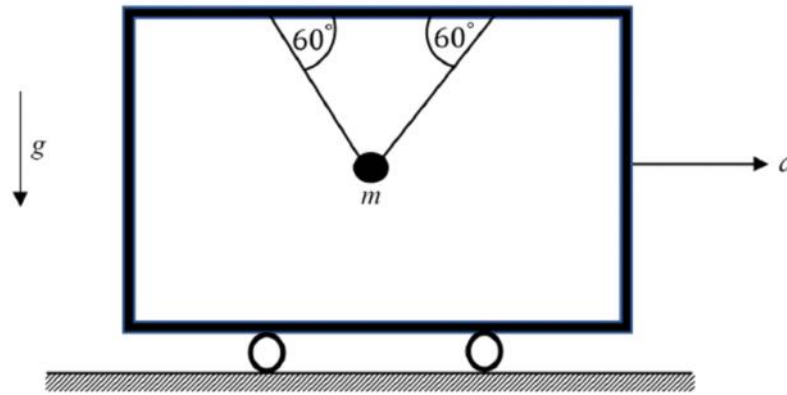
$$\therefore \omega = 2 \text{ rad/s}$$

$$F_c = m r \omega^2$$

$$= 5 \times 2 \times 2^2$$

$$\therefore F_c = 40 \text{ N.}$$

2M x E A steel ball of mass $m = 10 \text{ kg}$ is suspended from the ceiling of a moving carriage by two inextensible strings making 60° with the horizontal as shown. The carriage has an acceleration a such that the tension in the string on the right is double the tension in the string on the left. Take the acceleration due to gravity (g) as 10 m/s^2 . The acceleration a (in m/s^2) is (rounded off to one decimal place).



$$\begin{aligned}\sum F_y = 0 : T \sin 60^\circ + 2T \sin 60^\circ &= mg \\ \Rightarrow 3T \sin 60^\circ &= mg \\ \therefore T &= \frac{mg}{3 \sin 60^\circ}\end{aligned}$$

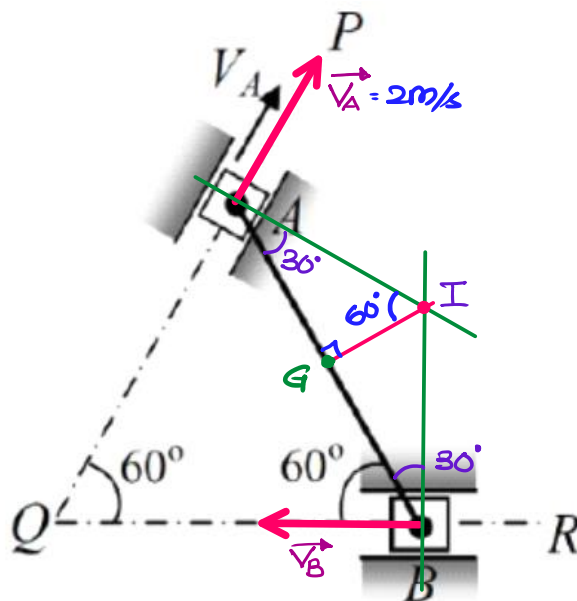
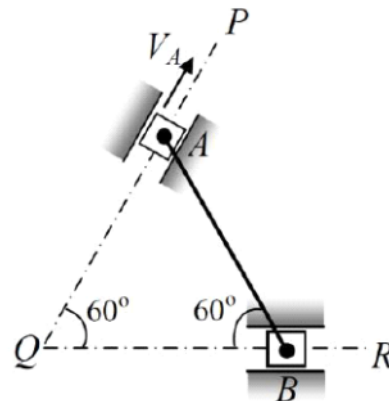
$$\begin{aligned}\sum F_x = 0 : 2T \cos 60^\circ &= T \cos 60^\circ + ma \\ \Rightarrow T \cos 60^\circ &= ma \\ \Rightarrow \frac{mg \cos 60^\circ}{3 \sin 60^\circ} &= ma\end{aligned}$$

$$a = \frac{g}{3 \cdot \tan 60^\circ} = \frac{10}{3\sqrt{3}} \text{ m/s}^2$$

$$\therefore a = 1.9245 \text{ m/s}^2$$

2M.

The rod AB , of length 1 m, shown in the figure is connected to two sliders at each end through pins. The sliders can slide along QP and QR . If the velocity V_A of the slider at A is 2 m/s, the velocity of the midpoint of the rod at this instant is _____ m/s.



$$V_B = \{IG\} \omega ; V_A = \{IA\} \omega$$

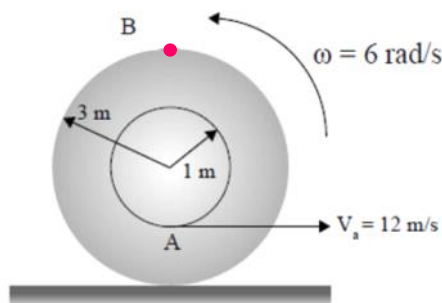
$$\Rightarrow \frac{V_B}{IG} = \frac{V_A}{IA}$$

$$\Rightarrow V_B = V_A \left\{ \frac{IG}{IA} \right\}$$

$$= V_A \left(\frac{IA \cos 60^\circ}{IA} \right)$$

$$\therefore V_B = 1 \text{ m/s}$$

The compound disk in the figure rolls on the horizontal floor. The linear velocity of the cord leaving A is 12 m/s rightward and the disk rotates counter-clockwise at 6 rad/s. The total velocity of the point B is



- (a) 4 m/s
(c) 16 m/s

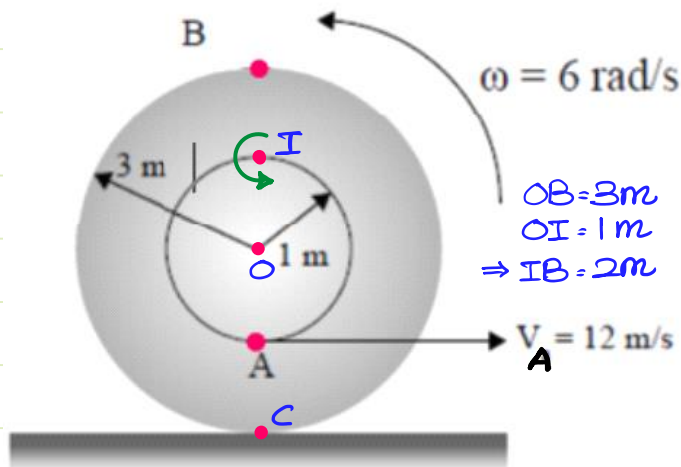
- (b) 12 m/s
(d) 20 m/s

Questions :

1. $\vec{V}_B = ?$

2. Identify if the disc is rolling with/without slip.

3. If it is rolling with slip find V_{slip} .



$$V_A = R_{A/I} \cdot \omega$$

$$12 = R_{A/I} \{6\} \Rightarrow R_{A/I} = 2m$$

There are 2 possibilities :

1. I Centre lies 2m above A.
2. I Centre lies 2m below A.

If I Centre lies 2m below A: $\Rightarrow V_A$ is in -ve x axis

But V_A is given in +ve x axis $\therefore$ I Centre lies 2m above A.
 $\therefore$ I Centre is not at pt. of contact $\therefore$ Rolling with slip.

$$V_B = R_{B/I} \cdot \omega$$

$$= 2m \times 6 \text{ rad/s}$$

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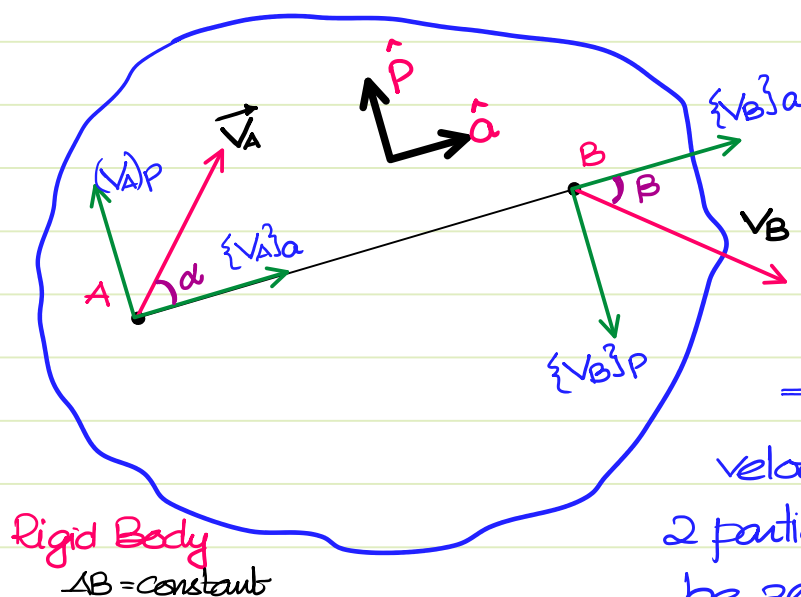
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 $\Rightarrow V_{\text{slip}} = V_{C/\text{surface}} = \vec{V}_C - \vec{V}_{\text{surface}}$

$$\therefore V_{\text{slip}} = V_C = R_{C/I} \cdot \omega$$

$$= 4 \times 6$$

$$\therefore V_{\text{slip}} = 24 \text{ m/s} \rightarrow$$

** Relative Velocity Method :

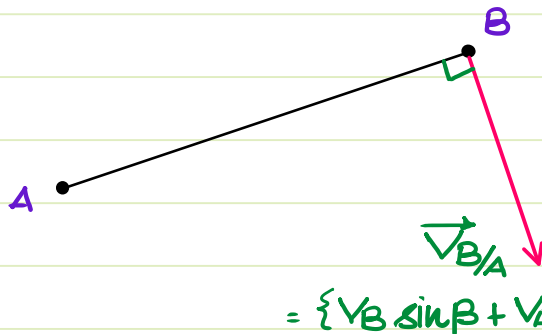


$\therefore$ the body is rigid,
 $\therefore |AB| = \text{constant}$
 $\Rightarrow (V_A)_{\text{along } AB} = (V_B)_{\text{along } AB}$

$$V_A \cos \alpha = V_B \cos \beta$$

$\Rightarrow$ The component of relative velocity along the line joining 2 particles in a rigid body must be zero.

⇒ The relative velocity must have only one component in the dirⁿ $\perp$ to the line joining 2 particles.



$$\vec{V}_B = (V_B \cos \beta) \hat{a} - (V_B \sin \beta) \hat{p}$$

$$\vec{V}_A = (V_A \cos \alpha) \hat{a} + (V_A \sin \alpha) \hat{p}$$

$$\vec{V}_{B/A} = \vec{V}_B - \vec{V}_A$$

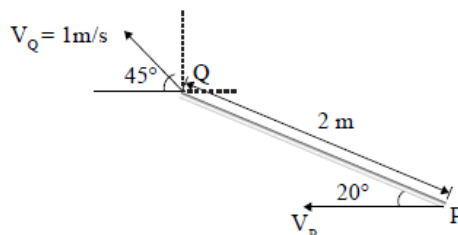
$$= \{V_B \cos \beta - V_A \sin \alpha\} \hat{a} + (-V_B \sin \beta - V_A \sin \alpha) \hat{p}$$

$$V_{B/A} = r_{B/A} \cdot \omega$$

Tips for solving problems:

1. Rel. Velocity along link = 0 $\Rightarrow V_B \cos \beta = V_A \cos \alpha$.
2. for angular velocity : $\omega = \frac{V_{B/A}}{r_{B/A}}$.

11. A rigid link PQ is 2 m long and oriented at 20° to the horizontal as shown in the figure below. The magnitude and direction of velocity V_Q and the direction velocity V_P are given. The magnitude of V_P (in m/s) at this instant is: (GATE-2014)



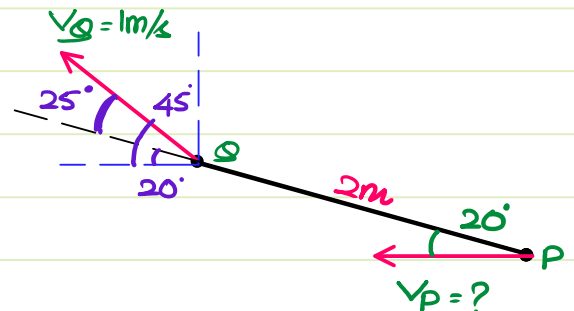
- (a) 2.14
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Find ω by Rel. Vel. Method.

$V_{P/Q}$ is $\perp$ to PQ

$$V_{P/Q} = V_P \sin 20^\circ + V_Q \sin 25^\circ = r_{P/Q} \cdot \omega$$

$$= 0.7525 = 2 \cdot \omega \Rightarrow \omega = 0.376 \text{ rad/s}$$



$\therefore PQ$ is rigid

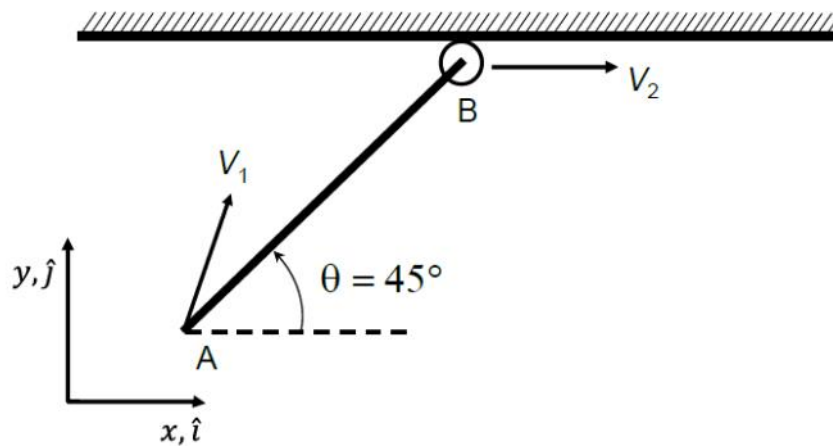
$\Rightarrow V_P$ along PQ = V_Q along PQ.

$$V_P \cos 20^\circ = V_Q \cos 25^\circ$$

$$\Rightarrow V_P = \frac{V_Q \cos 25^\circ}{\cos 20^\circ} = \frac{1 \times \cos 25^\circ}{\cos 20^\circ}$$

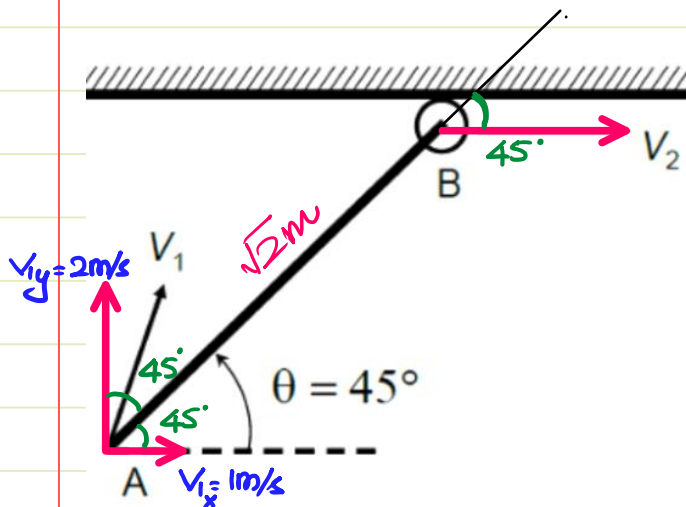
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The magnitude of the velocity V_2 (in m/s) at the end B is _____

$\therefore AB$ is rigid, $\therefore V_A$ along $AB = V_B$ along AB .



$$V_{1x} \cos 45^\circ + V_{1y} \sin 45^\circ = V_2 \cos 45^\circ$$

$$\Rightarrow V_2 = V_{1x} + V_{1y}$$

$$= 1 + 2$$

$$\therefore \boxed{V_2 = 3 \text{ m/s}}$$

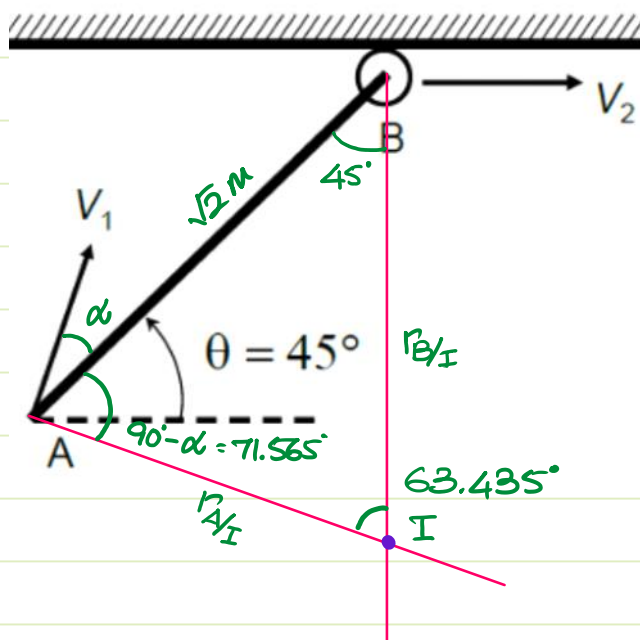
$$\underline{V}_B = \{3\hat{i} + 0\hat{j}\} \text{ m/s.}$$

$$\omega = ?$$

$$\omega = \frac{V_{AB}}{r_{AB}} = \frac{|\underline{V}_A - \underline{V}_B|}{\sqrt{2}} = \frac{|\hat{i} + 2\hat{j} - (3\hat{i} + 0\hat{j})|}{\sqrt{2}}$$

$$= \frac{|-2\hat{i} + 2\hat{j}|}{\sqrt{2}}$$

$$= \frac{2\sqrt{2}}{\sqrt{2}} \quad \therefore \boxed{\omega = 2 \text{ rad/s}}$$



$$\alpha = \tan^{-1} \left\{ \frac{V_{1y}}{V_{1x}} \right\} - 45^\circ$$

$$= \tan^{-1} 2 - 45^\circ$$

$$= 18.435^\circ$$

$$90^\circ - \alpha = 71.565^\circ$$

By sine rule

$$\frac{r_{A/I}}{\sin 45^\circ} = \frac{r_{B/I}}{\sin 71.565^\circ} = \frac{\sqrt{2}}{\sin 63.435^\circ}$$

$$r_{A/I} =$$

$$r_{B/I} =$$

$$V_A = r_{A/I} \cdot \omega$$

$$V_B = r_{B/I} \cdot \omega$$

4. Kinetics

4.1. Newtons' 2nd Law of Motion

4.1.1. Particle Study

* Newtons' 2nd Law of Motion

* D'Alemberts Principle.

* Centrifugal force

Newton's 2nd Law of Motion: When a non-zero unbalanced force acts on a body $\{\sum \vec{F} \neq 0\}$, then the body accelerates in the direction of unbalanced force with a Magnitude proportional to the force.

$$\sum \vec{F} \neq 0$$

$$\Rightarrow \sum \vec{F} \propto \vec{a}$$

$$\Rightarrow \boxed{\sum \vec{F} = m \cdot \vec{a}}$$

m : Mass of the body resists accⁿ, $\therefore$ it is the representative of Linear Inertia.

→ Newton's 2nd Law of Motion.

D'Alemberts' Principle: When a body experiences a non zero unbalanced force $\{\sum \vec{F} \neq 0\}$, then an imaginary force is considered in the dirⁿ opposite to that of acceleration in order to establish dynamic eqⁿ.

Note: This imaginary force is known as Inertia force & is always equal to $m \cdot a$.

By Newtons' 2nd Law: $\sum \vec{F} = m \vec{a}$

$$\Rightarrow \boxed{\sum \vec{F} + \{-m \vec{a}\} = 0} \quad \text{Dynamic Eqⁿ}$$

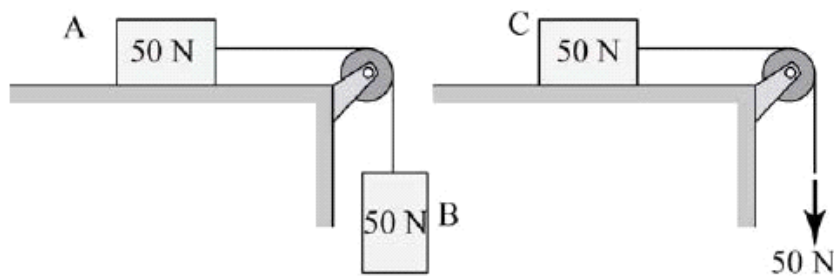
↓
Inertia force / Pseudoforce

If a body is undergoing a motion where the dirⁿ of $\vec{v}$ is changing then it must observe an accⁿ known as **Centripetal Accⁿ**.

The inertia force corresponding to Centripetal Accⁿ is known as **Centrifugal force** $\{ F_c = m r \omega^2 = m \frac{v^2}{r} \}$

F_c must be directed along the radius & away from the centre of curvature **{ Radially outward. }**

Two systems shown below start from rest. For the system shown on the left, two 50N blocks are connected by a cord. For the system shown on the right, the 50N block is pulled by a 50N downward force. Neglect friction. Which of the following is true?



- (A) Blocks A and C have the same acceleration.
~~(B)~~ Block C will have a larger acceleration than block A.
 (C) Block A will have a larger acceleration than block C.
 (D) Block A will not move.

Let a_1 be the accⁿ of block A & block B:

$$\sum F = m \cdot a_1$$

$$\Rightarrow 50 = \{m_A + m_B\} a_1 \Rightarrow 50 = \left\{ \frac{50}{g} + \frac{50}{g} \right\} a_1$$

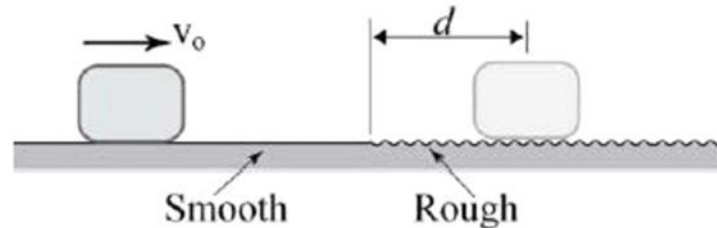
$$\Rightarrow a_1 = g/2$$

Let a_2 be the accⁿ of block C:

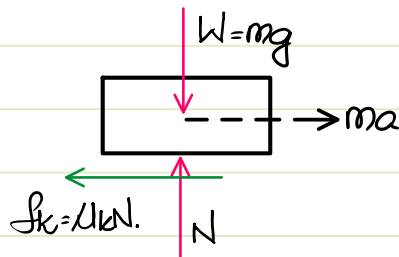
$$\sum F = m a_2$$

$$\Rightarrow 50 = m \cdot a_2 \Rightarrow 50 = \left\{ \frac{50}{g} \right\} a_2 \quad \therefore a_2 = g$$

- 2M. A block is travelling with a constant speed v_0 on a smooth surface when the surface suddenly becomes rough with a coefficient of friction μ , which causes the block to stop after a distance d . When the block travels twice as fast, i.e. at a speed $2v_0$ it travels a distance D on the rough surface before stopping. Find the ratio D/d .



Find the retardation on the rough surface:



$$\sum F_y = 0 : N = mg \Rightarrow f_k = \mu mg$$

$$\sum F_x = 0 : f_k = m \cdot a$$

$$\Rightarrow \mu mg = ma \therefore a = \mu g$$

When a body retards on a rough horizontal surface, without any other horizontal force, then $a = \mu g$.

$\therefore$ Retardation is constant $\therefore v^2 - u^2 = 2as$

$\Rightarrow s \propto \{v^2 - u^2\}$ if a is same in both cases.

$$\Rightarrow \frac{D}{d} = \frac{\{0^2 - (2v_0)^2\}}{\{0^2 - v_0^2\}} = \frac{-4v_0^2}{-v_0^2} \therefore \frac{D}{d} = 4$$

- 1M. A child of mass **5 kg** is going round a merry-go-round that makes 1 rotation in **3.14s**. The radius of the merry-go-round is **2 m**. The centrifugal force on the child will be

- (a) 40 N (b) 50 N
(c) 80 N (d) 100 N

$$\omega = 2\pi/t = 2\pi/3.14$$

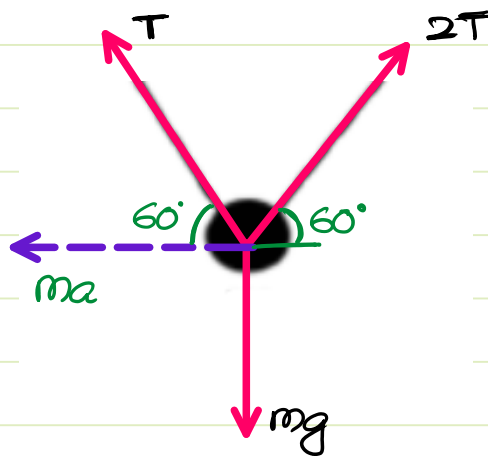
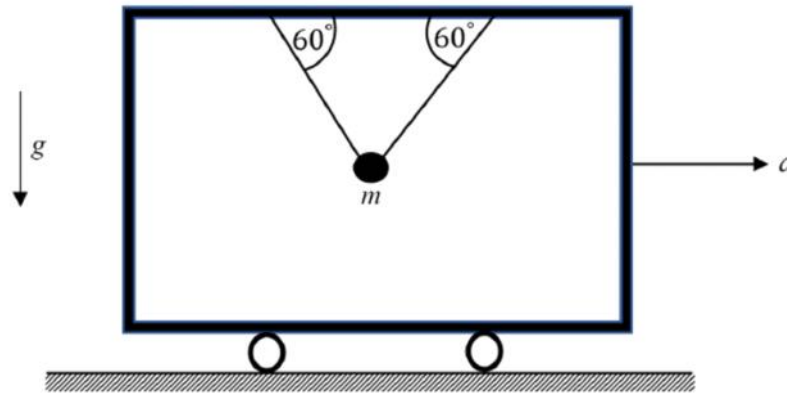
$$\therefore \omega = 2 \text{ rad/s}$$

$$F_c = m r \omega^2$$

$$= 5 \times 2 \times 2^2$$

$$\therefore F_c = 40 \text{ N.}$$

2M x E A steel ball of mass $m = 10 \text{ kg}$ is suspended from the ceiling of a moving carriage by two inextensible strings making 60° with the horizontal as shown. The carriage has an acceleration a such that the tension in the string on the right is double the tension in the string on the left. Take the acceleration due to gravity (g) as 10 m/s^2 . The acceleration a (in m/s^2) is (rounded off to one decimal place).

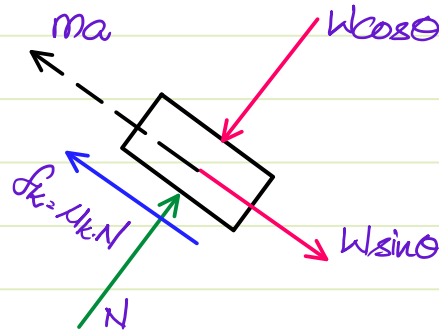


$$\begin{aligned}\sum F_y = 0 : T \sin 60^\circ + 2T \sin 60^\circ &= mg \\ \Rightarrow 3T \sin 60^\circ &= mg \\ \therefore T &= \frac{mg}{3 \sin 60^\circ}\end{aligned}$$

$$\begin{aligned}\sum F_x = 0 : 2T \cos 60^\circ &= T \cos 60^\circ + ma \\ \Rightarrow T \cos 60^\circ &= ma \\ \Rightarrow \frac{mg \cos 60^\circ}{3 \sin 60^\circ} &= ma\end{aligned}$$

$$a = \frac{g}{3 \cdot \tan 60^\circ} = \frac{10}{3\sqrt{3}} \text{ m/s}^2$$

$$\therefore a = 1.9245 \text{ m/s}^2$$



$$N = W \cos \theta \Rightarrow f_k = \mu_k W \cos \theta$$

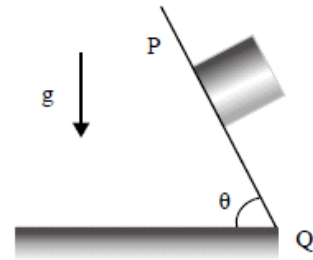
$$\cancel{g W \sin \theta} = \cancel{\mu_k W \cos \theta} + \cancel{ma}$$

$$\Rightarrow a = g \{ \sin \theta - \mu \cos \theta \}$$

$$s = \cancel{ut} + \frac{1}{2} at^2 \Rightarrow t = \sqrt{\frac{2s}{a}}$$

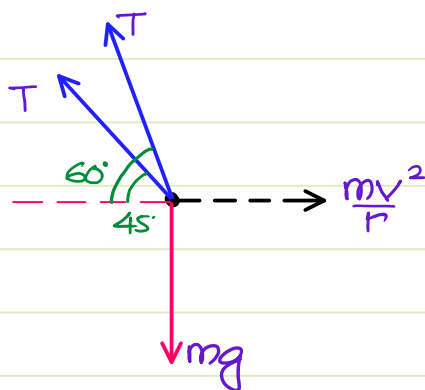
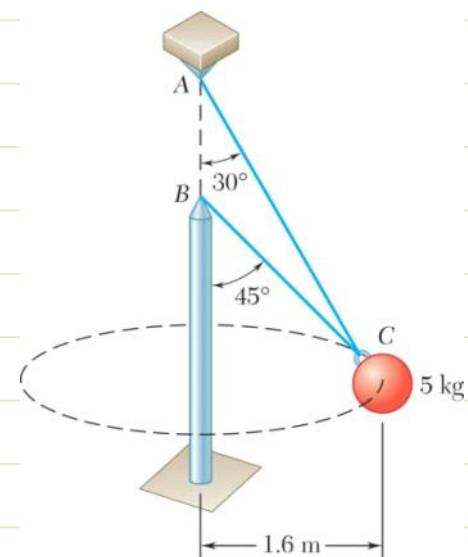
$$\Rightarrow t = \sqrt{\frac{2s}{g \cos \theta \{ \tan \theta - \mu \}}}$$

A block of mass M is released from point P on a rough inclined plane with inclination angle θ , as shown in the figure below. The coefficient of friction is μ . If $\mu < \tan \theta$, then the time taken by the block to reach another point Q on the inclined plane, where $PQ = S$, is (GATE - 2007)



- (a) $\sqrt{\frac{2s}{g \cos \theta (\tan \theta - \mu)}}$ (b) $\sqrt{\frac{2s}{g \cos \theta (\tan \theta + \mu)}}$
(c) $\sqrt{\frac{2s}{g \sin \theta (\tan \theta - \mu)}}$ (d) $\sqrt{\frac{2s}{g \sin \theta (\tan \theta + \mu)}}$

2M. A single wire ACB passes through a ring at C attached to a sphere that revolves at a constant speed v in the horizontal circle shown. Knowing that the tension is the same in both portions of the wire, determine the speed v .



$$\sum F_y = 0$$

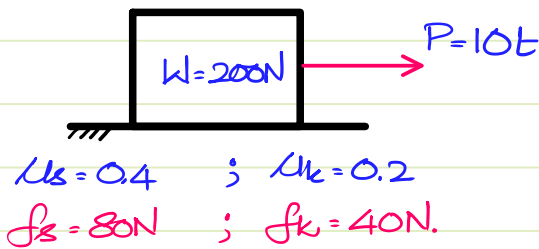
$$\Rightarrow T \sin 45^\circ + T \sin 60^\circ = mg$$

$$\Rightarrow T = \frac{mg}{\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2}} \Rightarrow T = 0.6356 mg$$

$$\sum F_x = 0$$

$$\Rightarrow \frac{mv^2}{r} = T \{ \cos 45^\circ + \cos 60^\circ \}$$

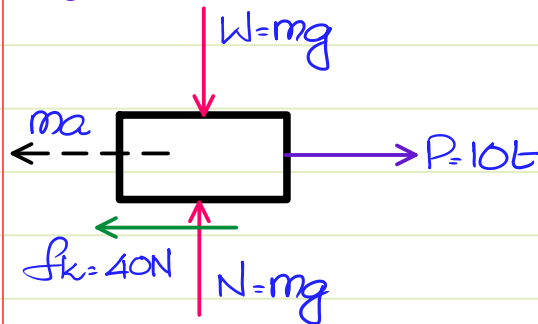
$$\Rightarrow v = \sqrt{0.7673 g r} \Rightarrow v = 3.47 \text{ m/s}$$



07. A block weighing 200 N is in contact with a level plane whose coefficients of static and kinetic friction are 0.4 and 0.2, respectively. The block is acted upon by a horizontal force (in newton) $P = 10t$, where t denotes the time in seconds. The velocity (in m/s) of the block attained in 10 seconds is _____. (GATE-2014)

For the body to be in dyn condition P must cross f_s
 $\therefore 10t = 80 \Rightarrow t = 8\text{s}$

After 8s:



$$\sum F_x = 0: 10t = 40 + m \left\{ \frac{dv}{dt} \right\}$$

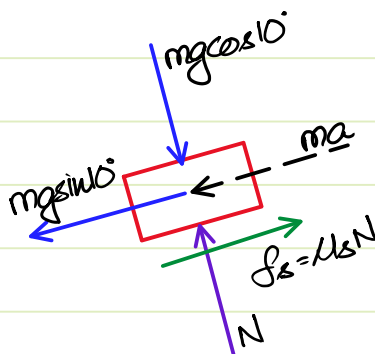
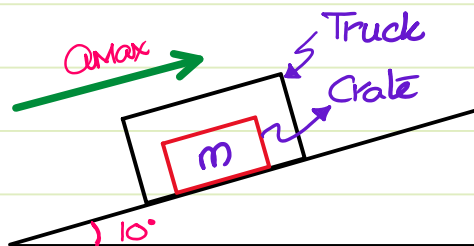
$$\Rightarrow \frac{dv}{dt} = \frac{1}{m} \{10t - 40\}$$

$$\Rightarrow \int_0^v dv = \frac{1}{m} \int_{t=8}^{t=10} \{10t - 40\} \cdot dt$$

$$v - 0 = \frac{g}{200} \{5t^2 - 40t\} \Big|_8^{10}$$

$$v = 4.905 \text{ m/s}$$

2M.



08. A truck accelerates up a 10° incline with a crate of 100 kg. Value of static coefficient of friction between the crate and the truck surface is 0.3. The maximum value of acceleration (in m/s^2) of the truck such that the crate does not slide down is _____. (GATE-2014)

$$N = mg \cos 10^\circ \Rightarrow f_s = \mu_s mg \cos 10^\circ$$

$$mg \sin 10^\circ + ma = \mu_s mg \cos 10^\circ$$

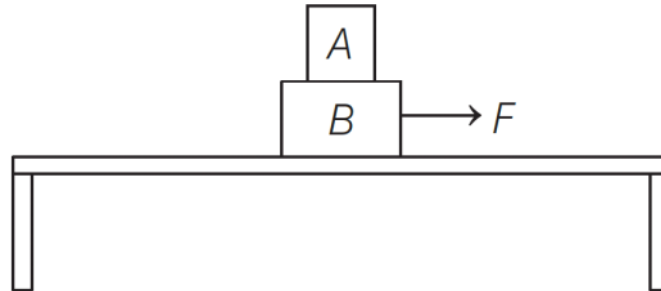
$$\Rightarrow a = g \{ \mu_s \cos 10^\circ - \sin 10^\circ \}$$

$$= 9.81 \{ 0.3 \cos 10^\circ - \sin 10^\circ \}$$

$$a_{\text{max}} = 1.194 \text{ m/s}^2$$

Two blocks A and B of masses $m_A = 1\text{kg}$ and $m_B = 3\text{kg}$ are kept on the table as shown in figure. The coefficient of friction between A and B is 0.2 and between B and the surface of the table is also 0.2 . The maximum force F that can be applied on B horizontally, so that the block A does not slide over the block B is
{Take, $g = 10\text{m/s}^2$ }

- (a) 12 N (b) 16 N (c) 8 N (d) 40 N



m_A :

$$N_1 = 10\text{N} ; f_1 = \mu N_1 = 2\text{N}$$

$$m_A a = f_1$$

$$\Rightarrow a = 2\text{m/s}^2$$

m_B :

$$N_2 = N_1 + 30 = 40\text{N}$$

$$\Rightarrow f_2 = \mu N_2 = 8\text{N}$$

$$F = f_1 + f_2 + m_B a$$

$$= 2 + 8 + 3\{2\}$$

$$F = 16\text{N}$$

